EARTHKEEPERS' BIOMASS SOLUTION INC.



Lignocellulose Geopolymer Bio-Aggregate Composite

SUSTAINABLE AND RENEWABLE

The Dawn of the New Era

PRODUCT DEVELOPMENT BEYOND

FIRE-RESISTANT

Alternative solutions play an important role in the housing construction industries. This, along with the rapid demand for the bio-aggregate products, represents a major potential market. It is sustainable at a price that will ensure sufficient supply of demand.

Summary:

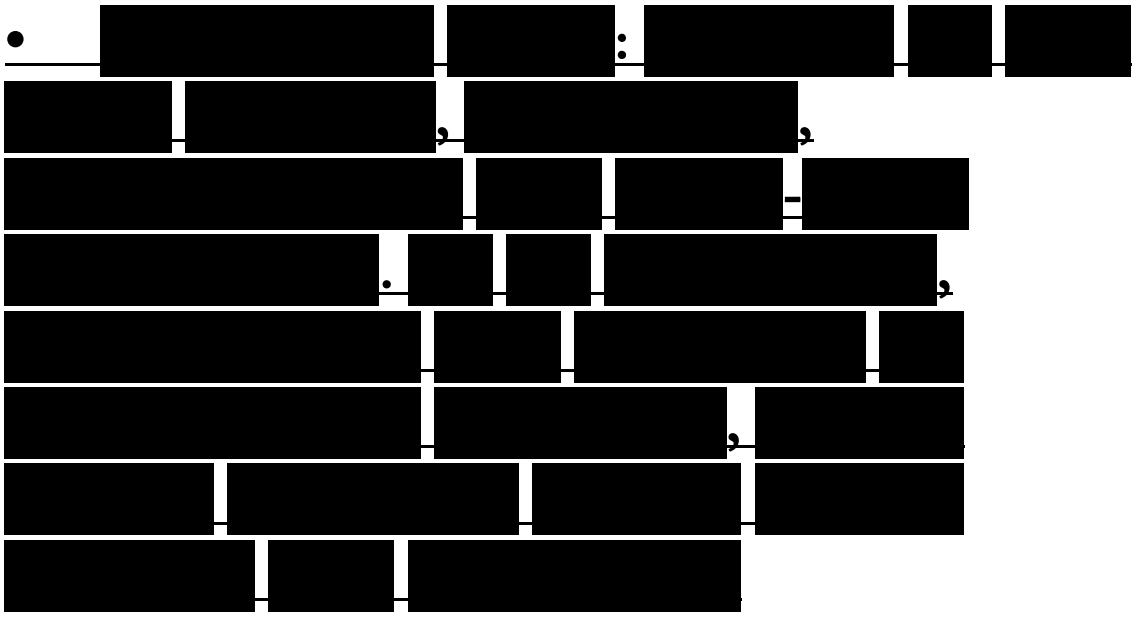
Earthkeepers' Biomass Solution Inc. (ESBI) presents innovative sustainable construction material called Lignocellulose Geopolymer (LCG) bio-aggregate composite. This product is designed beyond fire-resistant, renewable and eco-friendly, targeting large-scale construction projects, especially in wildfire prone and high temperature regions. The company manages a large tree plantation in the Philippines, cultivating a fast-growing species with high fiber content for pulp, and paper, and LCG composite production [1][2]

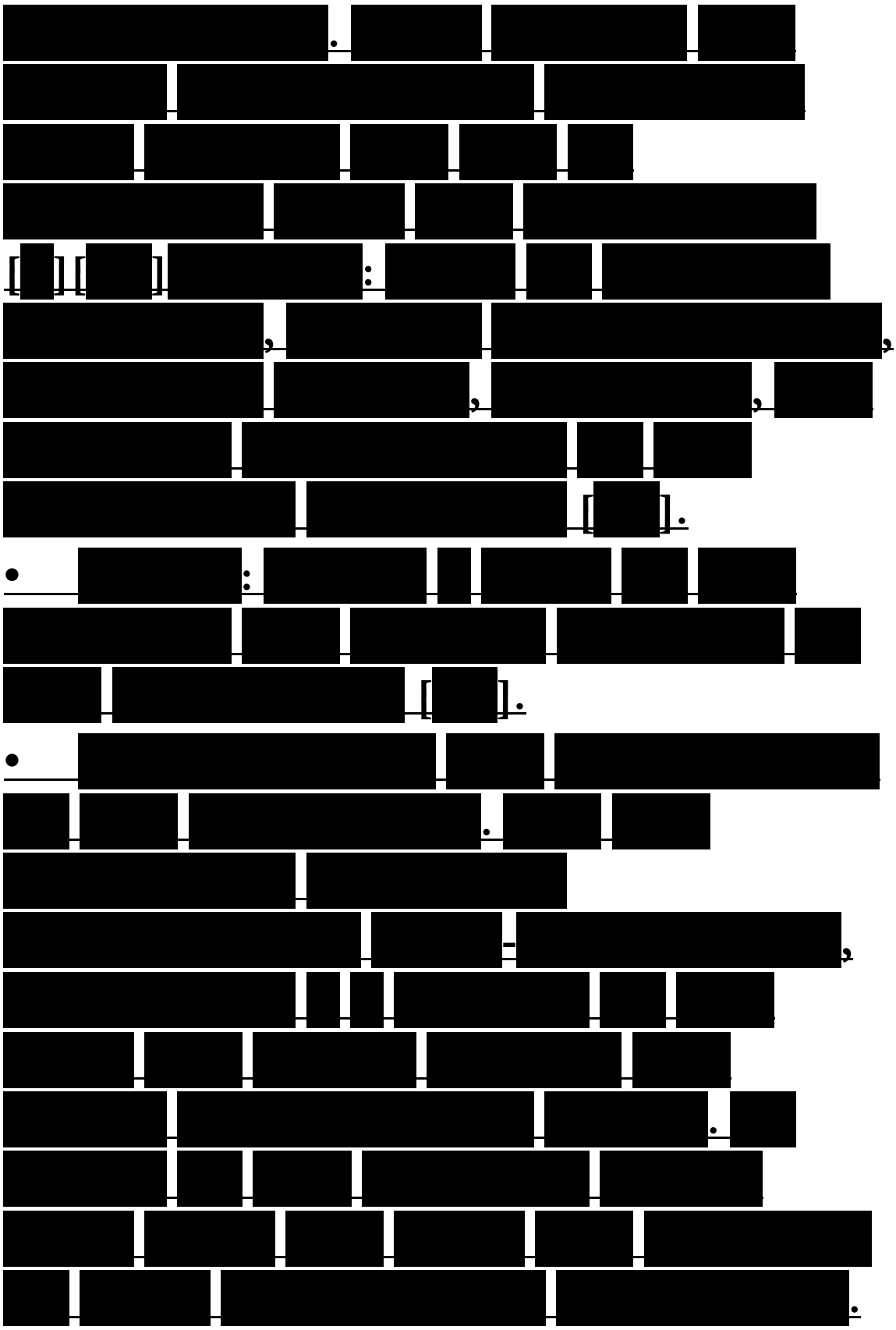
Company Overview and Vision:

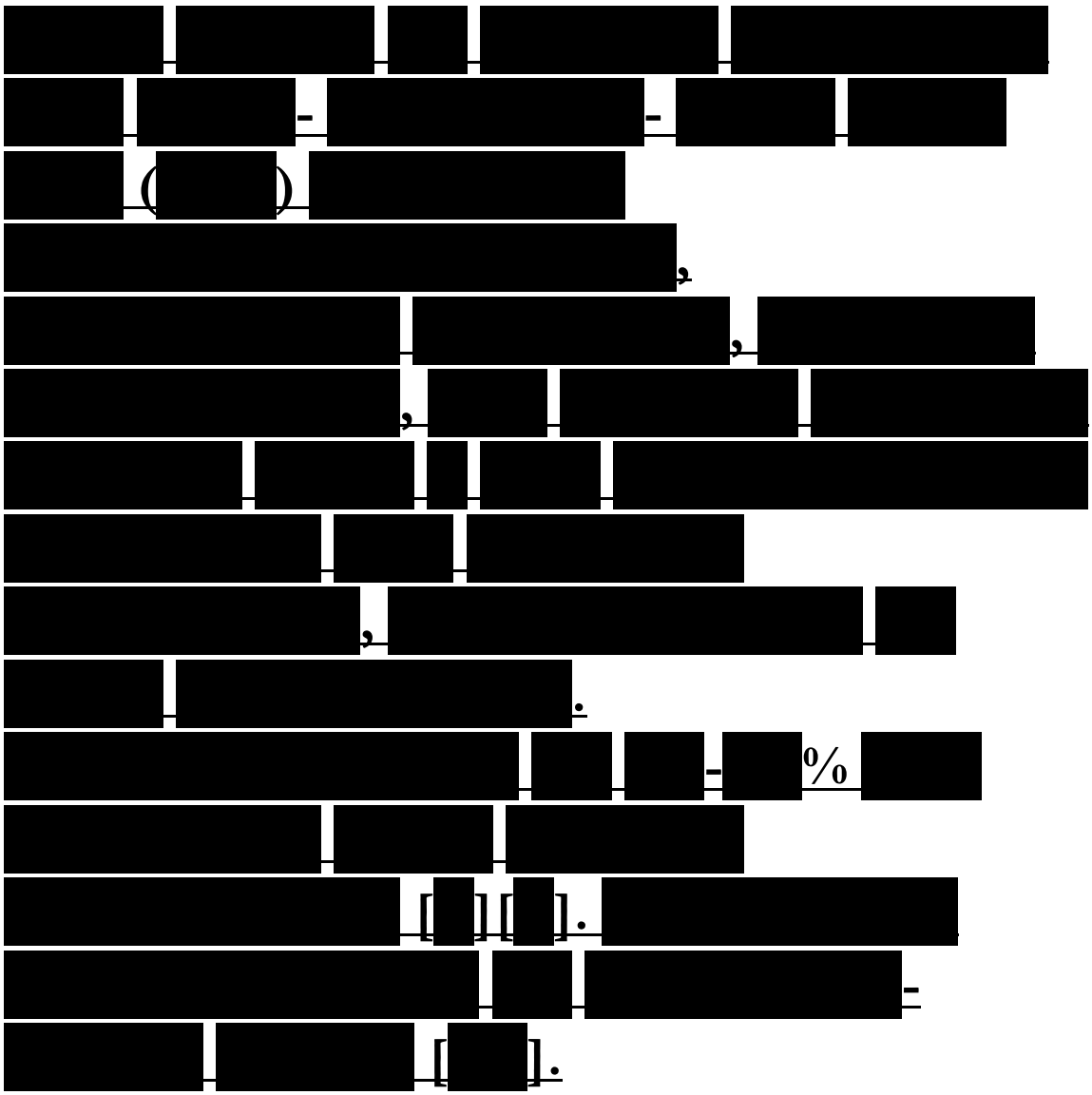
EBSI began operations in 1994 focusing on environmental afforestation and has evolved into a commercial entity with expertise in forestry, economics, chemistry, and engineering. The leadership includes CEO and founder Angelina Ferreras, along with consultants specializing in economics, pulp and paper, and electrical engineering. The company has pioneered environmental afforestation projects and

developed unique tree species for commercial use [3][4].

Key Components of Lignocellulose Geopolymer (LCG). The LCG bio-aggregate comprises five main elements.

- **Natural fibers:** Derived from the woody core of the fast-growing tree species, these lignocellulose fibers provide high tensile strength and thermal insulation. The biomass is mainly cellulose, hemicellulose, and lignin, which contribute to the composite's mechanical and ecological properties [5][6].
- carbon print. It acts as a supplementary
- 





- **Advantages of LCG Applications:**
  1. ***Fire-Resistance:*** Chars rather than burns, slowing fire-spread without chemical fireproofing [14].

2. **Seismic suitability:** Can be used in earthquake-resistant construction complying with relevant standards [15].
3. T**hermal Insulation:** Naturally regulates indoor temperature, reducing energy use for heating and cooling [16].
4. **Acoustic Insulation:** Provides excellent soundproofing due to its porous structure [17].
5. **Carbon Negative:** Absorbs more CO_2 and emits over its lifecycle, acting as a carbon sink [18][19].
6. **Impact Resistance:** Meets safety standards for impact-resistance on wall elements [20].
7. **Waterproofing**: Ensures watertightness of external walls elements under specified conditions [21].
8. **Hygrothermal Behavior**: Supports moisture regulation and summer comfort in buildings [22][23].
9. **Indoor Air Quality**: Porous nature traps air and regulates humidity, preventing mold and

improving air quality without chemical treatments [24][25].

10. **Fully Recyclable:** The material can be broken down and reused [26].
11. **Low Embodied Energy:** Minimal energy is used in manufacturing and transport [27].
12. **Versatility**: Suitable for a wide range of climates and building types, including load-bearing structural applications [28][29].

Comparison with Hempcrete and Other Materials: LCG geopolymer concrete blocks (GCB) offer advantages over hempcrete blocks in load capacity, durability, chemical resistance, and cost effectiveness. Both materials provide excellent insulation and sustainability benefits, but GCB is more suitable for structural applications [30][31].

Future Directions and Innovations:

EBSI invasions prefab LCG panels to reduce labor, cure time, and improve construction consistency. Integration of smart technology such as embedded sensors could enable real-time monitoring air quality,

humidity, and temperature in buildings made with LCG composite [32][33].

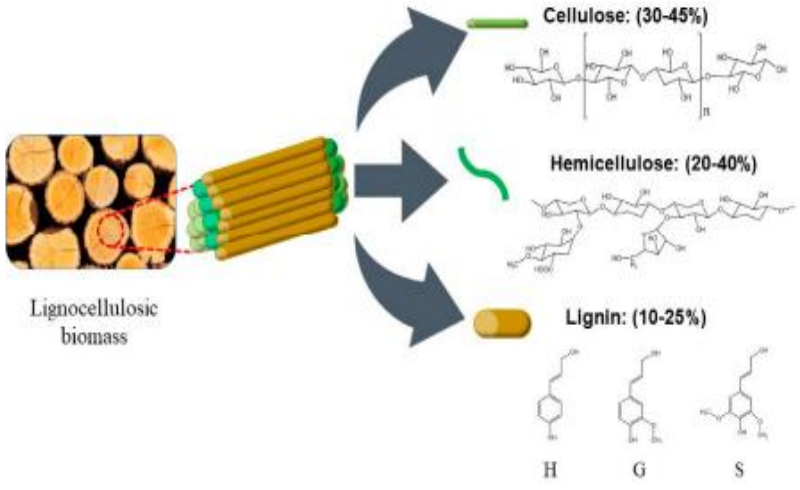
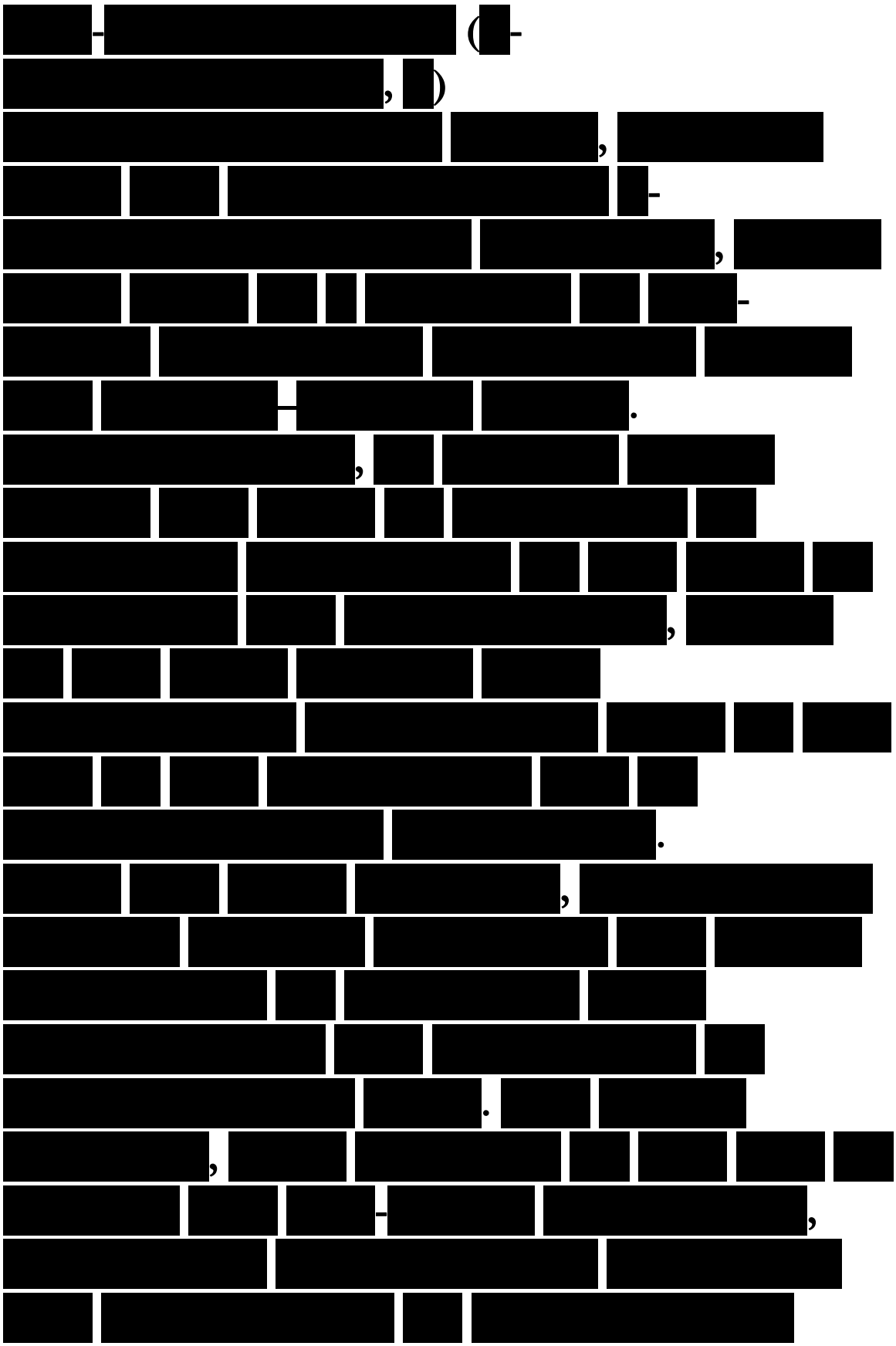


Figure 1

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[redacted]).Breathability: Due to high porosity and high permeability of lime, this assists the drying out of buildings and the avoidance of condensation problems, which greatly contributes to the comfort of people using the buildings.

Autogenous Healing: Rather than forming large cracks like concrete, due to expansion and contraction of the building, the small cracks that form are more likely to undergo self-healing aided by water penetrating these cracks and bringing the free lime up to the surface. When breaking concrete cylinders without fiber reinforcement, you will observe that the concrete breaks off in chucks.

Fibers are usually used in concrete to control cracking due to plastic shrinkage and drying shrinkage. They also reduce the permeability of concrete and thus reduce bleeding of

water. Some types of fibres produce greater impact-, abrasion, and shatter-resistance.

1. Water

Performance of Lignocellulose Geopolymer Bio-Aggregate Composite:

It performs exceptionally well in fire-test and earns a 0 rating on the Flame Spread and Smoke Development scale based on the standard test ASTM E84. While the blocks char on the surface when exposed to high heat for extended periods, it generally *does not create fire*, *combust*, or lose structural integrity.

Earthkeepers' Biomass Solution Inc., (EBSI)

Business Biographical Sketch & Partners

Overview:

Earthkeepers began operations in 1994 with environmental afforestation projects in the Philippines. Officially registered as EarthkeeperPI in 2002 in the United States (California) as a non-profit organization and in the Philippines with the Security & Exchange Commission (SEC). In 2017, the organization restructured into Earthkeepers' Biomass Solution Inc., as a stock corporation in the Philippines. Founded and led by Angelina Ferreras, the organization developed and pioneered environmental and commercial afforestation efforts, including the discovery of a distinct tree forest species in 1996.

Leadership and Key Personnel

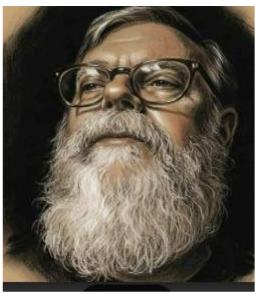


Angelina Ferreras, CEO & Founder

Educational Background: MBA in Business Management, UCSF

• Former Controller and General Manager of McKay & Company Hydraulics (Specialize in Marine Hydraulics Engineering systems). Managed operations, troubleshooting, repair, and maintenance.

• Led Earthkeepers in afforestation, research, and forest product development.



John R Fulper, Chemistry Division, Vice President

- Educational Background: B.SC in Professional Chemistry, Northern Illinois University

Professional Experience:

- Over 30 years in chromatography, filtration, and spectrophotometry, working with Pharma giants like Pfizer and General Electric (Cytiva).
- Role: Research and Product Development
- Publication
- Journal of Chromatography (1989): High-performance liquid chromatography method development.

Dr. Rodrigo G. Monterey Sr. – Operations Consultant

Educational Background:

- B.Sc. in Agriculture, M Sc. In Agronomy, Ph.D. in Horticulture, University of the Philippines.

• International Journal of Economic Research (2015): Rice production and self-sufficiency.

Professional Experience:

• Consultant and manager for agricultural and plantation projects globally, including Saudi Arabia and Guam.

• Professor Emeritus, University of the Philippines.

• Role: Advisor on agriculture operations, plantation management, and R&D since 2003.

Dr. Dante R. Garcia - Economic Productivity Consultant

• Educational Background:

• Ph.D. in Economics, University of Santo Tomas (UST).

• M.Sc. in Forestry, University of the Philippines.

• Professional Experience:

• Specializes in business economics, rural development, and marketing strategies.

• Published multiple studies on Philippine economic growth and productivity.

• Role: Develops strategies for non-wood forest product commercialization and community livelihood since 2014.

• Publications:

Ray N. Geganto - Technical Consultant for Pulp and Paper

• Educational Background: Chemical Engineering, University of the Philippines.

• Professional Experience:

• Consultant for CIBA-BASF, focusing on operational improvements in the paper and pulp industry.

• Expertise in waste management and environmental compliance.

• Role: Provides R&D support for paper production and alternative raw materials since 2013.

Marcos Tolentino - Electrical Engineering Consultant

• Educational Background: B.Sc. in Electrical Engineering, Mapua University.

• Professional Experience:

• Over 30 years of experience in designing electrical systems for industrial plants, petrochemical facilities, and nuclear test facilities.

• Significant global projects with Aramco, BP Alaska, and Chevron.

• Role: Designs systems for efficiency in biomass energy projects.

• Certifications and Training:

• Specialized training in AutoCAD and instrumentation design.

Earthkeepers' Key Accomplishments:

1. Pioneered research and development on unique tree species for environmental and commercial use.

2. Contributed significantly to environmental afforestation in the Philippines.

3. Assembled a team of globally recognized experts in forestry, economics, chemistry, and engineering.

EARTHKEEPERS' BIOMASS SOLUTION AFFORESTATION PROJECT



Figure 29 Afforestation farm in the province of Palawan



Environmental and Commercial Afforestation has been established beginning 1994



Figure 39, 40, 41 seedlings for the nursery establishment






7FEMS Reyes


YAMAHA







Figure 12 & 13



Figure 43 tree bark/bast fiber



Figure 44 heartwood



This bark was stripped off from the tree trunk. The bast fiber has a distinct feature of multiple fiber layers, unbroken throughout the entire length.

Bark Fiber, cleaned and cut one (1) meter long in **preparation** for decortication



Figure 45



Figure 46

This bark was stripped off from the tree trunk. The bast fiber has a distinct feature of multiple fiber layers, unbroken throughout the entire length. The bast fiber is for making pulp for paper etc..

Decorticating



Figure 47

Bast fiber



Figure 48

Pulp and



Figure 49

Paper



Figure 50



Figure 54 chipping processing



Figure 58 Wood Chips



Figure 59 Wood Pellets



alternative fuel against coal.





Lignocellulose Geopolymer Block

Figure 61 manufacturing machineries



Figure 62 employees



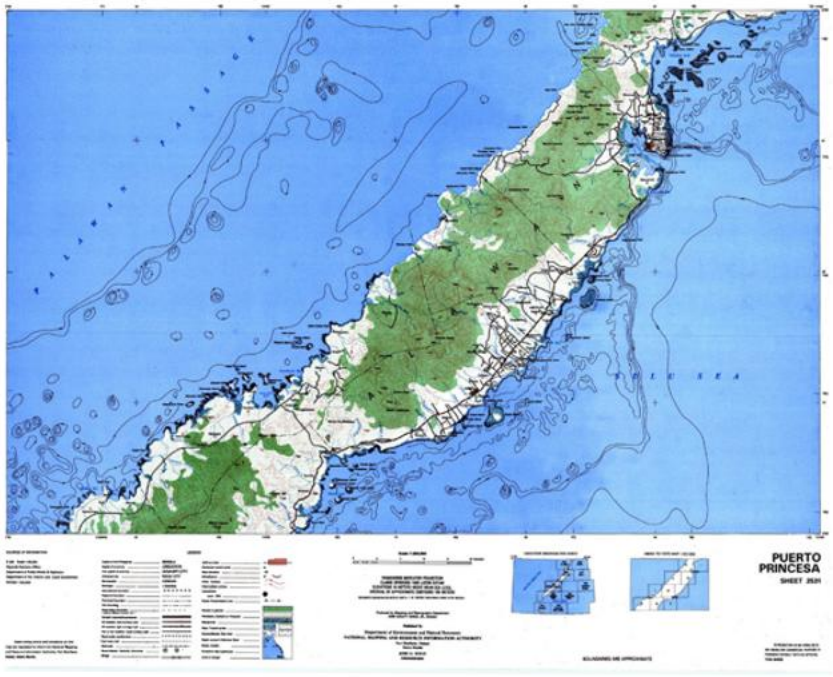

accuracy: 8m



SPECIAL QUALITY THAT SETS APART FROM OTHER FIBER PLANTS.

- Both the ***bast fiber and core fiber is*** highly oil absorbent species
- Have 2 distinct product of economic value: ***Bast Fiber and Core Fiber. Exceptionally high in Alpha Cellulose and Holocellulose.***
- Fast growing species, maturity is between 2-4 years old, depends on the soil fertility
  the tree height is about 45-60 feet, with 8-10 inches in diameter.
  Woody core; bone dry, 50-70 kilos per tree.
  Bast Fiber; air dry 3-5 kilos per tree.

- One hectare of land could be planted between 5,000 to maximum of 10,000 per hectare.
- It is not a monoculture
- It is a rain feed; therefore, it does not need irrigation.
- Fertilizer is not required.
- No herbicide or insecticide was used.
- Does not need mechanical equipment.
- Susceptive in in typhoon.



Palawan island is at the Southeast edge of the Philippines.
The length is 450 kilometers And has a width of 25-50 kilometers.

Shipping ports are in the Westside and Eastside of the island, dotted with shipping infrastructures, less than 10 kilometers from the shipping port.

Geopolymer Concrete Blocks (GCB) vs Hemp Crete blocks:

- **Insulation**: both geopolymer concrete blocks and hemp blocks offer excellent insulation properties and have the edge in moisture regulation and fire resistance.
- **Load bearing**: GCB can be used as a load-bearing material, making it more suitable for structural applications unlike hemp blocks, it is a non-bearing walls application and insulation.
- **Moisture Regulation**: Hempcrete excels in moisture regulation, as well as GCB, but in addition is more on durability and chemical resistance.
- **Sustainability**: both hempcrete and GCB are highly sustainable, and both renewable sources.
- **Environmental Impact**: is a low environmental impact, carbon-negative properties give it an edge in sustainability.
- **Cost-wise**: GCB is cost effective compared to Fer rock and Hemp Crete.

The Lignocellulose Geopolymer (LCG) in Sustainable Construction:

- Stronger, Smarter Binders that improve structural strength without losing its carbon-negative edge.

- Prefab LCG panels could change everything: Factory-made components would cut down on labor and curing time, speed up on-site construction, and improve consistency.

- Smart Tech Meets Natural Materials: Embedded sensors in LCG could monitor air quality, humidity, and temperature in real-tim

ADVANTAGES OF LCG APPLICATIONS:

1. **Fire-resistant-** chars rather than burns. That slows fire spread and provides valuable extra time in emergencies-without the need for added chemical fireproofing.
2. **Installation in seismic zones**- The process can be used for the construction of works requiring earthquake-resistant provisions within the meaning of the modified 2010 Order of October 22, provided that the requirements detailed in paragraphs 1.2.3.4.

3. **Thermal Insulation** –The walls naturally regulate indoor temperature. This reduces your reliance on heating and cooling systems-cutting energy bills and environmental load.
4. **Acoustics Insulation** – Natural insulation is excellent thermal and acoustic performance.
5. **Carbon**-**Negative**: it absorbs CO_2 than it emits over its lifecycle.
6. Impact Resistance - The process has been tested for safety impact resistance on wall elements in accordance with standard P 08-302. It meets the criteria for C2 and H2 classification, in accordance with CERIB test report nol027453
7. **Waterproofing of external walls –** The watertightness of the façade walls is adequately ensured, provided that the exposure conditions defined in article 4.2 of part 3 of the DTU are respected 20.1.
8. **Hygrothermal behavior –** The hygrothermal behavior of the wall is satisfactory, subject to compliance with the recommendations concerning the interior and exterior cladding.
9. **Summer comfort** – for the determination of the thermal inertia class of dwellings, which is an important factor for summer comfort, the external walls of this process belong to the category of distributed insulating walls.
10. **Indoor air quality-** Its porous, which means it traps air-great for thermal and sound insulation. It also absorbs and releases moisture, helping regulate humidity.
11. **Environmental Advantages:** LCG acts as a carbon sink It absorbs CO^2 as it grows, and even after processing, the final product stores more carbon than it emits. This gives it a net-negative carbon footprint.
12. **Fully Recyclable:** Break it down and reuse it – it is fully recyclable.
13. **Low Embodied Energy:** Minimal energy used in manufacturing and transport.
14. **Mold and Pest Resistant:** LCG properties naturally resist microbial growth. It regulates indoor humidity, prevents mold growth, and supports better air quality-without needing vapor barriers or chemical treatments.
15. **Carbon Negative:** It absorbs more CO_2 than it emits over its lifecycle.
16. **Soundproofing built in:** Its lightweight, porous structure dampens sound
17. **Versatility**: Is more versatile and can be used in a wider range of climates and building types.
18. **Load Bearing**: **Geopolymer concrete** can be used as a load-bearing material, making it more suitable for structural applications. It is more focused on durability and chemical resistance.



Young students travel in an overcrowded jeepney in suburban Palawan.



There can be no true democracy unless every citizen could share the wealth of the nation.

THANK YOU